VF Asia Footwear Laboratory Testing Manual

Test Method for Cardboard / Paperboard Adhesion	Test method # VF- F11 Page 1 of 3
Date Issued: May,2019 **Revision #: 1** **Reason for Revision: Rectify the unit of measure from kg to kgf for strength of material**	**Created by: Nila Babia Erazo**

PURPOSE AND SCOPE: To measure the level of adhesion between two or more layers of cardboards.

REFERENCE: ASTM D903

EQUIPMENT AND MATERIAL:

1. Tensile testing machine meeting the requirements of ASTM D638, preferably with graph recorder.
2. Clamping jaws for the tensile testing machine, which will not damage the materials being tested.
3. Calipers capable of measuring up to 0.01mm. Digital type is preferred.
4. Sharp cutter, straight and parallel-sided perpendicular to the board surfaces ***(Same cutter use for ECT)***
5. Beakers or Immersion Dish
6. Distilled water with a pH value of 6.5 ± 1.0
7. Stop-watch
8. Laminated cardboards or cartons with 2 or multiple layers

SAMPLE PREPARATION:

Condition the sample for at least 24 hours at the standard laboratory condition at 23.0±2.0 ° C Temperature and 55 ±10 % Relative Humidity.

1. **DRY TEST METHOD**
   Dimension of sample: Width @ 2.5 ± 0.1 cm and length @ 15 ± 0.1 cm.

2. **WET TEST METHOD**
   Dimension of sample: Width @ 5.0 +/-0.1cm and length @ 7.5 ± 0.1 cm.

3. **Use the cutter for ECT**

4. Inspect the edges of test specimens. Reject any that are not free from loose fibers on their edges or have one or more sides with non-parallel edges.

5. After cutting, the test piece edges should NOT be handled and must ensure the edges are not in contact with any other objects. No significant pressure should be placed on the specimens in any directions.

REPETITIONS: Prepare three (3) samples per direction.

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TEST PROCEDURE:

Test should be conducted at standard laboratory conditions @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

A. DRY TEST METHOD:

1. Measure the width of the test sample to the nearest 0.01cm. The measurement of width must be done before testing. (W).
2. Initiate at least 2 cm between the 2 layers of materials to be tested for the clamp to grip.
3. Zero and calibrate the tensile testing machine. Set the crosshead speed to 100 mm/minute and the jaw gap to approximately 3 cm.
4. Centrally clamp one side of the substrate of the bond sample into the stationary jaw of the tensile tester and the other side to the moving jaw. The sample must be inserted into the jaws such that the bond line can be observed by the operator as peeling progresses.
5. Operate the tensile testing machine at a speed of 100 mm/minute.
6. Continue testing until the bond sample has been completely separated.
7. Identify and record the maximum peak value on the chart recorder plot, to the nearest 0.1kgf. **(L)**
8. Calculate the force required to pull the test specimen apart by dividing the force by the measured width of the test specimen.

CALCULATION:

Adhesion Strength: kgf/cm

L = maximum peak in kgf

W = width of specimen in cm

TEST REPORT:

1. Adhesion strength - the maximum peaks to the nearest 0.1kgf/cm.
2. Other observation such as if there's tearing of specimens or peel- off. VF requires that lamination shall tear not peel-off. If there's no tearing it should be rejected.

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B. WET TEST METHOD.

1. Fill the glass beaker or immersion dish with distilled water
2. Soak the sample into the glass beaker for 5 minutes. Ensure that it is fully submerged in the water.
3. Take out the sample after 5 minutes
4. With the use of forceps, hold one side of the cardboard or paperboard at slightly slant position for 30 to 60 seconds.
5. Then check if the other side of the board delaminated.

EVALUATION:

After the specified time check visually if the cardboard or paperboard has delaminated. Sign of delamination should be rejected.

TEST REPORT: **Rating**:

- **PASS**: No delamination
- **FAIL**: With delamination